

Aim and Data

Research question: do HR, TEMP, and EDA features reveal experimental phase, or are they dominated by individual baselines and cohort effects?

- 312 observations = 26 individuals x 4 rounds x 3 phases.
- 51 physiological features from HR, TEMP, and EDA.
- Labels used only for validation: Phase, Individual, Cohort, Puzzler, Frustrated.

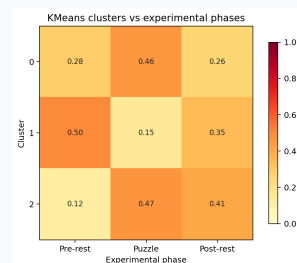
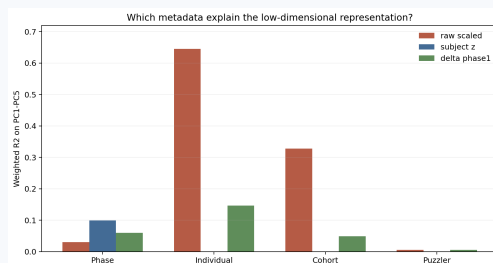
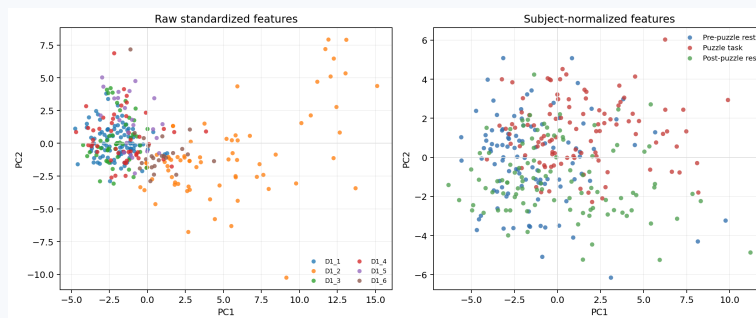
Pipeline

- Median imputation for the two missing feature values.
- Compare raw scaling, subject-wise z-score normalization, and phase1 baseline subtraction.
- PCA gives low-dimensional representations.
- KMeans with $k=3$ tests latent state discovery.
- LOSO logistic regression is used only as a supervised sanity check.

Key Numerical Results

- Raw PC1-PC5 explained variance: 58.4%.
- Raw representation R2: Individual 64.5%, Cohort 32.8%, Phase 3.0%.
- Subject-normalized phase R2 improves to 9.9%.
- KMeans phase agreement remains weak: ARI 0.075 after subject normalization.
- LOSO supervised sanity check: phase2-vs-phase1 AUC 0.779.

Evidence



Interpretation

Main conclusion: the raw biosignal representation mainly captures who was measured and which cohort the measurement came from, not a clean emotional state.

- Subject-wise normalization is essential for wearable biosignals.
- The phase signal is weak but nonzero.
- Unsupervised PCA plus KMeans does not robustly discover pre-rest, puzzle, and post-rest states.
- A negative result is valid here because the pipeline diagnoses baseline dependence and weak signal strength.
- Future work should use temporal features or self-supervised representations with subject-level validation.